

Ioannis Mavromatis

LEAD RESEARCH ENGINEER

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🎓 I. Mavromatis Scholar

At-a-glance

Lead Research Engineer with 10+ years' of R&D experience delivering cloud-native platforms and ML systems across large-scale distributed deployments. Proven track record in both research and development concepts, from feasibility and prototyping through to production-grade pilots in complex multi-cluster and cloud-edge environments. Strong focus on platform architecture, orchestration, observability, and ML/FL operations, with recent work on RAG, Agentic AI, and LLM-powered automation for deployment reasoning, capability discovery, and telemetry analysis.

Highlights

- **ML Systems:** MLOps/LLMOps, Federated & Distributed ML, model monitoring (drift/bias), energy-aware ML pipelines.
- **Cloud-Native Platforms:** Kubernetes (multi-cluster), microservices, GitOps packaging (Helm/kpt), API-driven automation.
- **Distributed Systems (Cloud-Edge Continuum):** orchestration, service placement, resilience/self-healing, scalability.
- **LLMs & Agentic AI:** RAG pipelines, tool-using agents for platform automation, LLM-assisted configuration/YAML generation, telemetry Q&A and self-healing.
- **Observability & Telemetry:** Prometheus/Grafana/Elastic, large-scale metrics/logs/traces, SLO-driven operations.
- **Networking & Systems Security:** secure AI pipelines, intrusion detection & prevention, anomaly detection, networking protocols.
- **Software:** end-to-end life-cycle management, strong development skills.
- **Technical leadership:** platform ownership, architecture decisions, mentoring/line management, stakeholder delivery.

Experience

Digital Catapult

LEAD RESEARCH ENGINEER

London, UK

Nov. 2023 - Present

Projects involved: – REASON / REASON+, SUSTAINET-guarDian, CHEDDAR CHEESE, SONIC, N-CONNECT.

- Leading the **architecture and technical delivery of AI-native orchestration platforms** for cross-domain cloud-edge infrastructures.
- Developing **distributed, cloud-native components** (microservices, APIs, etc.) for intelligent service orchestration and automation.
- Driving the **pre-productisation phase** of our in-house developed intelligent orchestration framework.
- Implementing **agentic AI pipelines** for platform automation, integrating LLM-based planners, tool routing, and structured APIs.
- Building **RAG frameworks over complex relational databases**, enabling natural language querying of platform capabilities and telemetry.
- Engineering **LLM-assisted Kubernetes and YAML generation agents** to automate deployment and QoS-aware configurations.
- Integrating **LLM co-pilots for observability and self-healing** across distributed cloud-native orchestration platforms.
- Designing **schema-aware semantic representations** of platform data to support scalable retrieval and reasoning.
- Developing **energy-efficient ML and Federated Learning pipelines** for resource-constrained cloud-edge environments.
- Performed a **large-scale study of the energy consumption** of Generative and Discriminative AI models.
- Building an **AI Verification, Validation, and Testing (VVT) framework** for ML-enabled distributed and autonomous platforms.
- Architecting **intelligent service placement and migration workflows** across heterogeneous domains (edge, cloud, multi-cluster).
- Driving end-to-end **platform PoCs** from system design to deployment, integration, and large-scale experimentation.
- Assessed the feasibility of **4G/5G nomadic and edge deployments** for connectivity in underserved regions.
- Authoring **bid proposals** across EPSRC, Innovate UK, and Horizon Europe funding bodies.
- Generating **IP and technical publications**.
- **Managing a team of 4 direct reports**, overseeing their career development and technical contributions.

Bristol Research & Innovation Laboratory, Toshiba Europe Ltd.

Bristol, UK

PRINCIPAL RESEARCH ENGINEER

Aug. 2019 - Nov. 2023

*Started as Research Engineer and progressed to Principal level.

Projects involved: – GreenML, UMBRELLA, SYNERGIA, CAVShield, CYTHEMIS.

- Architected and developed a **secure, scalable cloud-native IoT platform** supporting distributed devices and data pipelines.
- Designed **highly-available backend microservices and platform APIs** for large-scale IoT cloud-edge systems.
- Built **data drift detection and monitoring components** for long-running AI/ML systems.
- Researched and implemented **concept drift detection & mitigation strategies** for distributed ML deployments.
- Worked on **IID and non-IID data bias mitigation strategies** for FL training.
- Evaluated and deployed **ML and Federated Learning workloads** using Kubernetes-based MLOps workflows.
- Developed **energy-efficient ML optimisations** (quantisation, power capping, resource-aware training) for MLOps workflows.
- Implemented **federated identity and access management frameworks** for distributed IoT infrastructures.
- Led the development of a **cyber-secure platform** for Cooperative Intelligent Transportation Systems (C-ITS).
- Designed and implemented **distributed IoT testbeds and communication frameworks** for large-scale experimentation.
- Designed and developed an **air quality monitoring solution** for care homes.
- Worked on **secure synchronous flooding solutions** for Bluetooth communications.
- Worked on **end-to-end Intrusion Prevention Systems** evaluations.
- **Led a multidisciplinary team of software engineers** delivering platform-level microservices and system integrations.
- Provided **technical leadership and consultation** to industrial and governmental external partners.
- Led **WP 2 (CAVShield)** including technical strategy, reporting, and presentations to funding bodies and government stakeholders.
- **Mentored** interns, junior engineers, and researchers.
- Generated **IP and technical publications**.

University of Bristol

Bristol, UK

HONORARY RESEARCH ASSOCIATE

Aug. 2019 - Sep. 2023

- Migrated **AI-assisted Intelligent Transportation Solutions** to Cloud-native and Mobile Edge Infrastructures.
- Contributed to research on **distributed AI systems and edge-cloud continuum architectures**.

University of Bristol

Bristol, UK

RESEARCH ASSOCIATE (POST-DOC)

Oct. 2018 - Aug. 2019

Projects involved: – NG-CDI, FLOURISH.

- Developed an open-source **AI-assisted 5G simulator** for C-ITSs (DRIVE framework).
- Developed a **Fog Computing platform** for large-scale connected vehicle service experimentation.
- Implemented and deployed a large-scale **prototyped ITS-G5 testbed**.
- Conducted several **large-scale city-wide vehicular trials** campaigns.
- Designed an **active jammer** - tested performance/disruptions on prototyped ITS-G5 testbed.

University of Bristol

Bristol, UK

RESEARCH ASSISTANT

Feb. 2016 - Aug. 2018

Projects involved: – FLOURISH, VENTURER.

- Developed a **ITS-G5 vehicular communications PoC** platform.
- Conducted **city-wide connected vehicle trials** and analysed the experimental data.
- Developed a **parallelised** (multi-core compatible) **version of the OMNeT++ INET** framework.
- “Virtual Driving Testbed” - developed an **in-lab testbed emulation solution** for vehicular communication testing.

University Carlos III of Madrid (UC3M)

Madrid, Spain

NETWORK RESEARCH ASSISTANT (INTERN)

Apr. 2012 – Sep. 2012

Project involved: FLAVIA.

- Implemented **energy-efficiency optimisations** for **WLAN networks** (IEEE 802.11b/g/n/u).
- Modified **IEEE 802.11b/g drivers and firmware** (OpenWRT) to improve power consumption.
- Prototyped and evaluated a **custom wireless testbed** to assess system performance and energy impact.

NETWORKING AND IT POSITIONS

“Unifast” Kiriakoulis Georgios

Thessaloniki, Greece

IT AND NETWORK SUPPORT & CONSULTANT

Jul. – Oct. 2008, May – Aug. 2013

- Planned, provisioned and deployed **IT and networking infrastructure** (wired and wireless) for local businesses.
- Provided technical consultation on **ERP systems** and enterprise IT solutions.
- Delivered IT support across **Windows and Unix-based operating systems**.
- Installed and configured **satellite antenna systems** and related services.

TITAN Cement

Thessaloniki, Greece

IT AND NETWORK SUPPORT & CONSULTANT

Jul. 2007 – Aug. 2007

- Planned and deployed **IT and networking infrastructure solutions** for industrial environments.
- Programmed and tested **PLC and SCADA automation systems**.
- Provided IT support for **Windows and Unix-based systems**.

TEACHING POSITIONS

- 2017/2018 **"Mathematical Methods for Computer Scientists"**, Organised by: *University of Bristol*, BSc 1st year (80h)
"Networking Protocol Principles", Organised by: *University of Bristol*, BSc 3rd year / MSc (20h)
"Programming in C", Organised by: *University of Bristol*, MSc (70h)
"Object Oriented Programming", Organised by: *University of Bristol*, BSc 2nd year (45h)
- 2016/2017 **"Mathematical Methods for Computer Scientists"**, Organised by: *University of Bristol*, BSc 1st year (80h)
"Programming and Algorithms 2", Organised by: *University of Bristol*, BSc 1st year (45h)
- 2015/2016 **"Digital Signal Processing Systems"**, Organised by: *University of Bristol*, MSc (21h)
- 2014/2015 **"Digital Signal Processing Systems"**, Organised by: *University of Bristol*, MSc (21h)

Education

Dept. of Electrical & Electronic Engineering, University of Bristol

Bristol, UK

PHD IN COMMUNICATIONS

Oct. 2014 - Oct. 2018

- **PhD Title:** "5G Communication Framework for Smarter Autonomous Vehicles".
- **1st Year's Research Group Project:** "Localisation Of Strayed Targets (LOST)" (taught phase).

Dept. of Electrical & Electronic Engineering, University of Bristol

Bristol, UK

MSc IN "WIRELESS COMMUNICATIONS AND SIGNAL PROCESSING"

Sep. 2013 - Sep. 2014

- **MSc Thesis:** "Propagation Modelling using Iterative Physical Optics".

Dept. of Information Technology, ATEI of Thessaloniki

Thessaloniki, Greece

BSc IN "INFORMATION TECHNOLOGY ENGINEERING"

May. 2007 - Nov. 2011

- **BSc Thesis:** "Studying Quality of Experience (QoE) over Wireless Networks".

Skills

LLMs / RAG	LangChain, Transformers, vLLM, Ollama, exo, prompt/tool routing, vector search/embeddings, RAG evaluation.
ML/FL	PyTorch, NumPy, Flower FL, MLflow, KubeFlow.
Programming	Python (platform/ML/automation), Go (services), C/C++ (performance-critical systems).
Cloud/Edge	Kubernetes (K8s, K3s, MicroK8s), Helm/kpt, Docker, multi-cluster patterns, API gateways (Nginx/HAProxy).
Data	PostgreSQL, InfluxDB, Redis, MySQL; data pipelines for telemetry/analytics.
Observability	Prometheus, Grafana, Elastic, Kibana, Fluentd/FluentBit.
Automation/CI	Git, GitHub Actions/Jenkins, Infrastructure automation (TravisCI, Ansible).
Virtualisation	Proxmox, OpenStack, Google Cloud Platform (GCP), Amazon Web Services (AWS).
Networking	OAuth2, OpenVPN, Wireshark/Tcpdump, Nmap, FTP, LDAP.
Simulators	OMNeT++ (including Veins), SUMO, NS-2, NS-3, OPNET.
OSs	Linux-based Distributions (including kernel and driver programming), OS X, MS Windows.

Patents & Selected Publications

05/2025	A. Sanchez-Mompo, I. Mavromatis, P. Li, K. Katsaros, and A. Khan, "Green MLOps to Green GenOps: An Empirical Study of Energy Consumption in Discriminative and Generative AI Operations"	Information MDPI
12/2024	I. Mavromatis, S. De Feo, and A. Khan, "FLAME: Adaptive and Reactive Concept Drift Mitigation for Federated Learning Deployments"	EWSN 2024
07/2024	I. Mavromatis, K. Katsaros, and A. Khan, "Computing Within Limits: An Empirical Study of Energy Consumption in ML Training and Inference"	ICEST 2024
03/2024	J. Gao, I. Mavromatis, P. Li, P. Carnelli, and A. Khan, "Mitigating System Bias in Resource Constrained Asynchronous Federated Learning Systems"	IEEE PerCom 2024
11/2023	O. Belarbi, T. Spyridopoulos, E. Anthi, I. Mavromatis, P. Carnelli and A. Khan, "Federated Deep Learning for Intrusion Detection in IoT Networks"	IEEE GLOBECOM 2023
11/2023	I. Mavromatis, S. De Feo, P. Carnelli, R. J. Piechocki, and A. Khan, "FROST: Towards Energy-efficient AI-on-5G Platforms - A GPU Power Capping Evaluation"	IEEE CSCN 2023
11/2023	H. Frijj, I. Mavromatis, A. Sanchez-Mompo, P. Carnelli, A. Olivereau, and A. Khan, "Multi-stage Attack Detection and Prediction Using Graph Neural Networks: An IoT Feasibility Study"	IEEE TrustCom 2023
06/2023	T. Chow, U. Raza, I. Mavromatis, and A. Khan, "FLARE: Detection and Mitigation of Concept Drift for Federated Learning based IoT Deployments"	IEEE IWCMC 2023

	I. Mavromatis, A. Sanchez-Mompo, F. Raimondo, J. Pope, M. Bullo, I. Weeks, V. Kumar, P. Carnelli, G. Oikonomou, T. Spyridopoulos, and A. Khan, "LE3D: A Lightweight Ensemble Framework of Data Drift Detectors for Resource-Constrained Devices"	IEEE CCNC 2023
01/2023		
10/2022	C. Lockie, I. Mavromatis, A. Stanoev, Y. Jin, and G. Oikonomou, "Securing Synchronous Flooding Communications: An Atomic-SDN Implementation"	EWSN 2022
04/2022	I. Mavromatis, A. Stanoev, A. J. Portelli, C. Lockie, M. Ammann, Y. Jin, and M. Sooriyabandara, "Reliable IoT Firmware Updates: A Large-scale Mesh Network Performance Investigation"	IEEE WCNC 2022
03/2022	US449398, "System and Method for Detecting and Rectifying Concept Drift in Federated Learning"	U.S. Patent
09/2019	X. Wang, I. Mavromatis, A. Tassi, R. Santos-Rodriguez, and R. J. Piechocki, "Location Anomalies Detection for Connected and Autonomous Vehicles"	IEEE CAVS 2019
04/2019	A. Tassi, I. Mavromatis, R. J. Piechocki, and A. Nix, "Secure Data Offloading Strategy for Connected and Autonomous Vehicles"	IEEE VTC 2019

Honors & Awards

09/2024	AI Award for High Tech & Telecom, for REASON project , National AI Awards, <i>Software/Network Architect</i> The National AI Awards was launched to spotlight the remarkable advances and contributions in the field of artificial intelligence.
09/2022	Industrial Innovation Award, for UMBRELLA project , Connected Britain, <i>Lead Backend Architect / Developer</i> Industrial Innovation Award is given to projects making an important contribution in stimulating technology development, business start-ups and growth.
05/2019	IEEE Best Paper Award , IEEE Vehicular Technology Conference (IEEE VTC-Spring) Best Paper award for my work "Agile Data Offloading over Novel Fog Computing Infrastructure for CAVs".
12/2018	IEEE Popularity Award , IEEE Vehicular Networking Conference (IEEE VNC) Popularity award for my work "Parallel Implementation of the OMNeT++ INET Framework for V2X Communications".

Grants & Proposals

09/2025	CHEDDAR CHEESE , EPSRC, <i>Solution Lead, WP Lead, Bid Writer</i> CHEDDAR CHEESE will address orchestration issues across the cloud-edge computing continuum of future platform solutions, will investigate how Agentic AI can enable auto-configuration and self-healing, and how explainability will enhance users' and system administrators' experience. <i>Project is led by Kings College London</i>	£200,000
04/2025	REASON+ , DSIT, <i>Tech Lead, Bid Writer</i> REASON-6G develops a roadmap for open 6G network architectures. <i>Project is led by Smart Internet Lab - University of Bristol</i>	£1,500,000
07/2020	SYNERGIA , Innovate UK, Grant no. 53707, <i>Bid Writer</i> SYNERGIA devised, developed and demonstrated a novel secure-by-design, endpoint-to-core IoT platform for large-scale networks of low-power resource-constrained devices. <i>Project is led by BRIL - Toshiba Europe Ltd.</i>	£2,177,373
01/2020	CAVShield , Innovate UK, Grant no. 133989, <i>WP Lead, Bid Writer</i> The CAVShield project addressed the urgent global need for robust methods and techniques to identify and measure cyber-vulnerabilities in fleets of connected vehicles. <i>Project was led by Honda R&D Europe (U.K.) Ltd.</i>	£427,480

Supervision & Mentoring

PHD & RESEARCH SECONDMENTS

2022–2023 (12 months)	Graham Peden Subject: Malicious Firmware Binary Code Vulnerability Detection. Co-advisors: Prof A. Rashid, Dr A. Khan.	<i>PhD in Cybersecurity</i> University of Bristol
2022–2023 (12 months)	Othmane Belarbi Subject: Concept Drift Detection and Mitigation using Serverless Computing. Co-advisors: Dr T. Spyridopoulos, Dr E. Anthi, Dr A. Khan, Dr P. Carnelli.	<i>PhD in Cybersecurity</i> University of Cardiff

2022–2023 (12 months)	Vasilis Ieropoulos Subject: Machine Learning Threat Detection for Low Complexity Edge Deployment: Threats, Risks and Mitigations. Co-advisors: Dr T. Spyridopoulos, Dr E. Anthi, Dr A. Khan, Dr P. Carnelli.	PhD in Cybersecurity University of Cardiff
2023 (6 months)	Hamdi Friji Subject: ML-based Intrusion Detection System: A Graph Neural Network Approach. Co-advisors: Dr P. Carnelli, A. Sanchez Mompou, Dr A. Khan.	PhD in Green Cybersecurity CEA
2018–2019 (11 months)	Jonathan Thomas Subject: AI-assisted Communication Framework for Self-Driving Vehicles. Co-advisors: Prof R. Piechocki, Dr R. Santos-Rodriguez.	PhD in Communications University of Bristol

RESEARCH INTERNS

2023 (6 months)	Stefano De Feo Subject: Federated Learning Concept Drift Detection and Mitigation: A Large-Scale Evaluation. Co-advisors: Dr P. Carnelli, Dr A. Khan.	Industrial Internship, BSc Toshiba Europe Ltd
2021 (3 months)	Charles Lockie Subject: Reliable IoT Firmware Updates: A Large-Scale Mesh Network Performance Investigation. Co-advisors: A. Stanoev, Dr Y. Jin.	Industrial Internship, BSc Toshiba Europe Ltd
2020 (3 months)	Jonathan Hebditch Subject: Privacy and Anonymity of an Industrial IoT Platform: A VPN Approach.	Research Internship Toshiba Europe Ltd
2020 (6 months)	Iosif Koen Subject: Enabling a Smart Edge Computing Plane for CAVs with Kubernetes. Co-advisors: Prof R. Piechocki, Prof P. Chatzimisios.	Erasmus+ Research Internship International Hellenic University
2019 (6 months)	Samual Taylor Subject: A PiCar-Based Collision Avoidance Scenario: Design Insights and Experimental Testbed. Co-advisors: Prof R. Piechocki, Dr A. Tassi.	Undergrad. Research Internship University of Bristol
2018 (6 months)	Panagis Artoumas Subject: Enabling Full-Stack City-Scale Simulations for Connected and Autonomous Vehicles. Co-advisors: Prof R. Piechocki, Prof P. Chatzimisios.	Erasmus+ Research Internship International Hellenic University

MEng/MSc STUDENTS

2023 (6 months)	Hanyue Zhang Subject: Drift Detection Framework for Multi-Dimension Data Streams. Co-advisors: Dr A. Khan, Dr P. Carnelli, Dr P. Li.	Industrial MSc Project Toshiba Europe Ltd
2023 (6 months)	Jikun Gao Subject: Towards Federated Learning at Scale: A System Bias Investigation. Co-advisors: Dr A. Khan, Dr P. Carnelli, Dr P. Li.	Industrial MSc Project Toshiba Europe Ltd
2023 (6 months)	Shengzhe Huang Subject: Deep Reinforcement Learning-Based Efficient Task Allocation in Federated Learning. Co-advisors: Dr A. Khan, Dr P. Carnelli, Dr P. Li.	Industrial MSc Project Toshiba Europe Ltd
2023 (6 months)	Vincent Wong Subject: Quantisation and Federated Learning: Improving Sustainability through Energy Reduction. Co-advisors: Dr A. Khan, Dr P. Carnelli.	Industrial MEng Project Toshiba Europe Ltd
2023 (6 months)	Stefano De Feo Subject: GreenML: A Smart Way Forward for Sustainable Machine Learning. Co-advisors: Dr A. Khan, Dr P. Carnelli.	Industrial MEng Project Toshiba Europe Ltd
2022 (6 months)	Charles Lockie Subject: Cybersecurity over Large-Scale Mesh Networks with Synchronous Flooding. Co-advisors: A. Stanoev, Dr Y. Jin.	Industrial MEng Project Toshiba Europe Ltd

2021 (6 months)	Marius Ammann Subject: Everything under one UMBRELLA: Large-Scale Experimentation with a Multi-Radio IoT Testbed. Co-advisors: A. Stanoev, Dr A. Portelli.	<i>Industrial MEng Project</i> Toshiba Europe Ltd
2021 (6 months)	Prithvi Jaisingh Subject: Distributed Intrusion Detection using Novel Federated Learning Approaches. Co-advisors: Dr P. Carnelli, Dr T. Spyridopoulos, Dr A. Khan.	<i>Industrial MEng Project</i> Toshiba Europe Ltd
2020 (6 months)	Jianwen Zhu Subject: A Low Latency MEC Solution for 5G Network Slicing. Co-advisors: Dr J. Thota, B. Holden.	<i>Industrial MSc Project</i> Toshiba Europe Ltd
2020 (6 months)	Sanyam Vays Subject: AI-Based Intrusion Detection System for Industrial IoT. Co-advisors: Dr T. Spyridopoulos, Dr A. Khan, Dr P. Carnelli.	<i>Industrial MEng Project</i> Toshiba Europe Ltd
2020 (6 months)	Philip Masters Subject: Wireless Coordination of F1/10 Autonomous Vehicles. Co-advisors: Dr A. Portelli, Dr M. Baddeley.	<i>Industrial MEng Project</i> Toshiba Europe Ltd
2019 (5 months)	Marie Aylward Subject: Benefits of Connected and Automated Vehicles on Large-Scale Urban Traffic Flow Use-Cases. Co-advisors: Prof R. Piechocki.	<i>BSc Project</i> University of Bristol
2019 (5 months)	Maria Fernanda Espino Gonzalez Subject: Spectrum Sharing Strategies for WiFi and Intelligent Transport Systems (ITS). Co-advisors: Prof R. Piechocki.	<i>MSc Project</i> University of Bristol
2019 (5 months)	Xinluan Bai Subject: Semi-Persistent Scheduling for C-V2X Vehicle-to-Vehicle Communication. Co-advisors: Prof R. Piechocki.	<i>MSc Project</i> University of Bristol
2019 (5 months)	Yuanri (Blues) Xing Subject: Design Practices on an Agile Communication Infrastructure for Autonomous PiCars. Co-advisors: Prof R. Piechocki.	<i>MSc Project</i> University of Bristol
2019 (5 months)	Yudi Fan Subject: A PiCar-Based Collision Avoidance Scenario: Sensor Data Acquisition and a Simulated Approach. Co-advisors: Prof R. Piechocki, Dr A. Tassi.	<i>MSc Project</i> University of Bristol
2019 (5 months)	Zhe (Anton) Zhao Subject: On the Enhancement of Traffic Flows Sharing CAVs Trajectory Intention. Co-advisors: Prof R. Piechocki, Dr A. Tassi.	<i>MSc Project</i> University of Bristol
2018 (6 months)	Matthew Uppington Subject: Anomaly Detection Mechanisms for Cooperative Autonomous Vehicles. Co-advisors: Prof R. Piechocki.	<i>BSc Project</i> University of Bristol
2016 (5 months)	Yu (Teresa) Bi Subject: Acquisition and Broadcasting of Enhanced Vehicle Positioning Information via a Linux-Based System. Co-advisors: Prof R. Piechocki, Dr A. Tassi.	<i>MSc Project</i> University of Bristol
2016 (5 months)	Xiaoyu Ye Subject: Reducing the Interference of Vehicular Communications. Co-advisors: Prof R. Piechocki, Dr A. Tassi.	<i>MSc Project</i> University of Bristol

Scientific Activities - Involvement

CONFERENCE / WORKSHOP ORGANISATION

- 2025 2nd Workshop on Enabling Machine Learning Operations for next-Gen Embedded Wireless Networked Devices (EMERGE) - at *International Conference on Embedded Wireless Systems and Networks (EWSN)*
- 2024 1st Workshop on Enabling Machine Learning Operations for next-Gen Embedded Wireless Networked Devices (EMERGE) - at *International Conference on Embedded Wireless Systems and Networks (EWSN)*

TPC MEMBER

- 2016 - IEE ICC, IEEE PIMRC, IEEE VTC, IEEE IFIP, IEEE CSCN, IEEE VNC, IEEE DIPI, ACM EWSN, CPS-IoTBench, Ad-Hoc Now,
2025 IARIA VEHICULAR, and more.

EXTERNAL REVIEWER

- Journal IEEE Communications Magazine, IEEE Transactions on Vehicular Technology, IEEE Wireless Communications Magazine, IEEE Journal On Selected Areas in Communications, IEEE Access, Springer Wireless Networks, Elsevier Vehicular Communications, Elsevier Computer Networks, MDPI Sensors, MDPI Applied Sciences, Data in Brief, and more
- Conf. IEEE ICC, IEEE GLOBECOM, IEEE CSCN, IEEE VTC, IEEE VNC, IEEE PIMRC, IEEE IFIP, ACM EWSN, IARIA VEHICULAR, Ad-Hoc Now and more.

EXTERNAL JUDGE

- 2019 Google Science Fair - Preliminary Judge & Scientific Accuracy Judge
2016 Google Science Fair - Preliminary Judge

Invited Talks & Panel Discussions

- 07/2024 **"6G Technologies Roadmap"**, Organised by: *IEEE 6G Summit*, Panel Discussion
- 01/2023 **"Role of Machine Learning in Next Generation Wireless Networks"**, Organised by: *IEEE Consumer Communications & Networking Conference*, Panel Discussion
- 03/2022 **"Industrial R&D: Career and Opportunity Perspectives"**, Organised by: *School of Computer Science and Informatics, Cardiff University*, Invited talk

Training & Certificates

- 08/2020 **Kubernetes MasterClass : Kubernetes & Docker Swarm for DevOp**, Udemy
- 08/2020 **Machine Learning**, Coursera, organised by Stanford University

Volunteering

- Technical Program Committee member and reviewer - several conferences and journals (2017 - now).
- Industrial Mentoring Scheme, Faculty of Engineering, University of Bristol (2019 - 2022).